CivicTriage: Multilingual Civic Grievance Triage

Product Requirements Document | Version 1.0 | Academic project (B.Tech final year)

1. Overview

CivicTriage is a lightweight web app for municipal offices that receive large volumes of citizen grievances (water, roads, sanitation, street lights, noise, encroachment) in different languages. It reads each grievance, works out how upset the citizen is and how quickly it needs action, and places it in the right queue so staff handle the most pressing items first.

The focus is a small, reliable core: understand the text, rate emotion and priority, show a ranked queue. Everything else is deliberately left out of version 1.

2. Problem Statement

Municipal helplines and web forms collect grievances as plain text. Staff read them in arrival order, so a burst water main or an unsafe open manhole can wait behind a routine request. Grievances in regional languages are often read late or skipped. There is no quick way to see which complaints come from distressed citizens or which areas keep complaining.

3. Goals and Non-Goals

Goals

- Accept grievances in English, Hindi and Telugu (typed text only).
- Assign each grievance an emotion, a priority level and a civic category.
- Show staff a ranked queue with a short plain-language reason for each ranking.
- Keep all records so nothing is lost on refresh or restart.

Non-goals for version 1

- No voice input, file or photo uploads.
- No model fine-tuning; classification uses a prompted local LLM only.
- No user-management console, OTP flows or role hierarchy beyond two roles.

- No semantic search over past cases, PDF reports or mobile app.
- No automatic assignment to individual staff; routing stops at the department queue.

4. Users

Persona	Need	Key actions
Citizen	Report a problem simply and see progress	Submit grievance, view own status
Ward officer	Know what to handle first	View ranked queue, change status, add a note

Only these two roles exist. Officer accounts are created manually in the database by the developer.

5. Scope Summary

Area	Version 1	Deliberately excluded
Languages	English, Hindi, Telugu	Other languages, code-mixed handling
Emotion labels	Angry, Anxious, Sad, Neutral	Fine-grained emotions
Priority levels	Immediate, Soon, Routine	Numeric priority score
Categories	Water, Roads, Sanitation, Electricity, Other	Legal or criminal categories
Classifier	Prompted LLM via Ollama	Transformer training, model comparison
Interface	Single web app, two role views	Chat assistant, voice, maps

6. Functional Requirements

FR1 Grievance submission. A citizen enters text, picks an optional ward name, and submits. The record is saved immediately with status Received.

FR2 Language handling. The system detects the language. Hindi and Telugu text is sent to the LLM with an instruction to understand it and reply in English labels. No separate translation model is used.

FR3 Classification. For each grievance the LLM returns emotion, priority, category and a one to two sentence reason as structured JSON. If the output is invalid, retry once, then mark the record Needs Review.

FR4 Priority rules. Priority combines the LLM result with a short keyword list (for example: flooding, live wire, accident, no water for days). A keyword hit can raise priority one level but never lower it.

FR5 Officer queue. Officers see a table sorted by priority, then by age. Filters: priority, category, ward, status. Each row shows the reason text.

FR6 Status updates. Officers can set Received, In Progress or Resolved and add a short note. Citizens see the latest status and note.

FR7 Basic insights. One page with three charts: grievances per category, grievances per priority, grievances per ward. Data export as CSV.

FR8 Authentication. Email and password sign-in for citizens and officers. Citizens can self-register; officers cannot.

7. Data Model

Table	Main fields
users	id, name, email, password hash, role (citizen or officer)
grievances	id, user id, ward, original text, detected language, english summary, emotion, priority, category, reason, status, created at
status_notes	id, grievance id, officer id, note, created at

All data lives in a single relational database that persists across restarts.

8. Screens

1. Sign in and register
2. Citizen home: submit form and list of own grievances with status
3. Officer queue: ranked table with filters and detail drawer
4. Insights: three charts and CSV export button

9. Suggested Technology (flexible)

- Interface and backend: Streamlit or Flask
- Database: SQLite or PostgreSQL
- LLM: any small model served locally through Ollama
- Language detection: langdetect
- Charts: Plotly or Matplotlib

The implementer may substitute equivalents as long as the functional requirements are met.

10. Non-Functional Requirements

- Classification result shown within 15 seconds of submission on a normal laptop.
- Passwords stored hashed; citizens can only see their own grievances.
- A failed classification never deletes or hides a grievance.
- Interface usable on a phone-sized screen.

11. Success Metrics

- At least 80 percent agreement with hand-labelled priority on a 60-sample test set (20 per language).
- Fewer than 5 percent of grievances ending in Needs Review.
- Officer queue loads in under 3 seconds with 500 records.
- Demo covers the full path: submit, classify, rank, update status, view insights.

12. Milestones

Week	Deliverable
1	Requirements, database schema, sample dataset of 60 labelled grievances
2	Sign in, registration, grievance submission and storage
3	LLM classification with JSON handling and keyword rules
4	Officer queue, filters and status updates
5	Insights page, CSV export, accuracy check on test set
6	Testing, documentation and demo preparation

13. Risks

Risk	Mitigation
LLM misreads Telugu or Hindi	Test with the labelled set early; add examples to the prompt
Slow responses on weak hardware	Use a smaller model; classify in the background
Inconsistent labels	Write a one-page labelling guide before creating the dataset

14. Open Questions

- Which wards or zones should be preloaded?
- Should citizens be allowed to submit anonymously?
- Is Telugu the right third language for the target region?